nsl: a Newspeak dialect for Zephyr — what it keeps, what it drops, and why

and the case for an agent-first OO language

DRAFT 2, 2026-10-07, for iteration with David. Sources are bracketed keys, e.g. [VM], resolved in the References; in this Markdown they are links. The PDF beside this file is rendered from it.

Abstract

Embedded VMs used to live or die by porting: many RTOSes, each with its own drivers, network stack and security story, and a VM had to follow them all. Zephyr's reach changes that: a VM ported once can reach more than 900 boards, with connectivity, security and drivers built in — Thread among them, more work than a small project could build alone. Zephyr's cost is the learning curve, which its own community names its most persistent challenge [ZT10].

nsl is a Newspeak dialect that puts an approachable object-capability language over Zephyr. Zephyr carries the freight, nsl directs traffic: the radio, the crypto and the drivers stay in C, and the interpreter pays mostly for decisions. A host compiler produces a read-only image that a small C VM runs in place from flash, in the roughly 112 KB that Zephyr, OpenThread and mbedTLS leave on a 256 KB nRF52840. nsl keeps Newspeak's structure — modules as objects, no globals, capabilities as the unit of authority — and drops most of its dynamism, each drop with its reason and its cost.

Last, the case for an agent-first OO language. Agents can't smell trouble, but they can find patterns in data, so every tool answers in data: the whole language fits in a context window, every development verb is a machine-readable HTTP call, every profile sample or crash names a source line, and the loop is short — the host VM first, then a device load, not a reflash. Zephyr's contributors don't trust AI-generated code without heavy review [ZT10]; nsl bets that a fast, repeatable loop supplements that review.

1. The constraints

nsl's floor is a 256 KB-RAM microcontroller with a radio. That is a hard target but not an extreme one: it is about as small as a part can be and still hold Zephyr, a Thread stack and an application, and it covers common radio parts, the nRF52840 and the ESP32-C6 among them. Smaller parts are not a goal. Size was the target, but simplicity was the mechanism: the July 2026 rewrite removed non-local return, fibers and growable stacks because they had become "a cascade of interacting complexity … with the VM ballooning far past its simple intent" [Pivot], and the bytes it saved were modest — 3 921 B of code, 676 B of RAM [MA]. Most of the choices in §4 would hold on a large board too (Figure 1).

What 256 KB leaves. Measured from the linker symbols of the nRF52840 dongle's Thread build (2026-10-03):

Use	RAM
VM arena: objects, the VM's own 16 KB stack, floats	112 KB
Zephyr kernel, drivers, USB, thread stacks	57 KB
Zephyr system heap	32 KB
OpenThread and the 802.15.4 radio driver	25 KB
Network buffers	14 KB
mbedTLS	10 KB
Spare	4 KB

Zephyr and its radio stack take about 140 KB and nsl gets the rest, with 4 KB to spare at link time. The VM's code is not the tight axis — about 42 KB of a 1 MB flash. RAM is.

The tightest arena is not on the smallest part. Across the boards the VM runs on [Boards]:

Board	RAM	VM arena	What sets it
ESP32 (WROOM)	520 KB, of which Zephyr maps ~192 KB as data RAM	48 KB	WiFi's driver heap and TCP buffers take the rest
ESP32-S2 (MagTag)	320 KB	64 KB	WiFi; 92 KB links, 64 KB is what has booted
nRF52840	256 KB	112 KB	Zephyr, OpenThread and mbedTLS; 4 KB spare
ESP32-C6	512 KB	128 KB	a chosen default with margin; 256 KB links with the radio

WiFi, not the size of the part, is what squeezes the classic ESP32 to 48 KB.

An interpreter, with no JIT. There is no room here for a JIT's code cache, so every method runs in the interpreter. One method activation costs about 12 µs (11.83 on the C6, 11.13 on the RP2350 [RP2350]), roughly a hundred times the same work in C. That is cheap at the rate events arrive — a timer, a packet, a button — and expensive per byte, which is why the radio, the crypto and the drivers stay in C, and why code that turns out hot can follow them (§2).

A program that cannot be written at run time. The program is a read-only image in its own flash partition, executed in place [Image]. The firmware is flashed once; programs are loaded beside it.

A peer that is not nsl. A gateway written in Go mirrors each node's state over UDP and CBOR, so the wire format is a contract between two languages, not an nsl object serialisation [Porta].

2. The VM

oevm is a register VM in C, built on Piumarta and Warth's Open Extensible Object Model [OEOM]: object, symbol and vtable are the only primitive kinds, and everything else — arithmetic included — is a message that a vtable answers. Instructions are 32 bits and fixed length, 31 of them; small integers take a fast path and fall back to a send. One instruction carries Newspeak's implicit-receiver send, which looks up a name in the lexically enclosing classes before the superclasses. The interpreter is one flat, non-recursive loop over one fixed-capacity stack, and a context lives on the heap so that a block can outlive the frame that made it — which is the whole reactive model (§3). Memory is a fixed arena with a precise mark-sweep collector and no `malloc` once running [VM].

The kernel is nsl, not C: collections, the value algebra, the CBOR codec, channels, the space, timers, the supervisor and the gateway bridge are all nsl source. C carries data, nsl carries policy.

Binding a driver: no FFI, one primitive. nsl code cannot call an arbitrary C function. A device is reached through a *primitive*: a selector on a closed list, 160 of them today, each backed by one C function that calls the Zephyr driver API. Binding a new driver means appending one name to that list, writing the C function, and adding the kernel method that hands the device out — through `platform` — as a capability. The change is additive: nothing already on the list moves, and no app can add a primitive of its own [Serial]. The firmware's author extends the machine; the program only uses what it was given.

Native code replaces nsl code without anyone noticing. Because every operation is a send through a vtable, a method is a binding from a selector to a closure, and whether that closure is interpreted or native is one field. To move a hot method to C, write the C and bind it: the callers, the interpreter and the image format do not change. A spike did this for the Towers of Hanoi on an ESP32, and the native method ran 116–119 times faster than the interpreted one at two problem sizes [Hanoi]. That is a best-case ceiling — all the work in one method, the recursion fully native — not a steady-state figure for a whole program. The point is the mechanism: a VM that dispatches on type tags would need a special case on every call path; this one needs one binding.

3. A reactor, not threads

Zephyr is multithreaded; nsl is not. The VM runs on one Zephyr thread, and the C side — OpenThread, the network stack, the liveness observer — runs in threads around it. Inside nsl there is one reaction at a time, and each runs to completion.

State lives in a **space**: a tree of paths whose leaves are *cells* (the latest value) or *streams* (append-only). A program registers a reaction with `whenever:do:`; a write to a path, a timer firing or a packet arriving queues the reactions that care, and the reactor runs them one by one. Timers post from their interrupt to a mailbox; the reactor blocks on it when there is nothing to do [VM].

Why not threads, the usual answer on an RTOS:

- **No locks and no data races in nsl code.** A reaction cannot be interrupted by another, so shared state needs no protection and an agent writing a reaction cannot get it wrong.
- **RAM is predictable.** Threads cost a stack each, sized for the worst case. The reactor has one stack, and the rewrite that removed fibers removed their per-fiber stacks with them [MA].
- **Errors are contained, not fatal.** A reaction that fails is caught by the supervisor, which keeps the heap and the reactor running and restarts or quarantines the job [MD].

The cost is the one every event loop has: a long computation starves the timers. The answer is to split it across reactions, and the guard is the **liveness observer**, a C thread that judges the reactor's progress against the shortest timer period the program has declared. A wedged node is dumped, rebooted, and leaves a note saying why [Liveness].

Always-on and deep sleep. A node's power mode is a value at a path, written by the gateway and acknowledged by the node — not a command [Porta]. An *always-on* node runs a loop that never returns; it is the mains profile. A *deep-sleep* node wakes, takes its reading, exchanges with the gateway and sleeps again; `platform deepSleep:` never returns, and the next wake is a full reboot. Its program is declared *run-once*, because a loop that expected to keep running would be killed by each sleep. Deep sleep runs on the MagTag today. What makes it work is that every exchange is started by the node: a sleeping node has no address to be called on, so the gateway holds what it wants — a new setting, a new program — until the node next wakes and asks.

4. The dialect against Newspeak

What drives each nsl divergence from Newspeak

Numbers are rows of NEWSPEAK_DIALECT.md. Filled = the primary reason the record gives; ring = a secondary one.

		RAM / flash budget	JIT-less interpreter	Capability security	Read-only image	Flat engine, run to completion	One wire meaning	Simplicity (YAGNI)
Static classes, no mixins, hand accessors	1, 2, 6		●					○
Bare Array/String; byte strings, streams	3, 10, 12	●						○
Dictionary on parallel arrays	11	●						○
No globals; platform; confined devices	8, 9, 13, 17, 32			●				
Reflection and debugger behind a mirror	14, 15, 16			●				
Control flow lowered to bytecode	19		○			●		
No NLR, no fibers, fixed stack	21, 22, 23	○				○		●
Errors are Records; predicate catch	7, 24					○	●	
Closed immutable value algebra	28						●	
One bounded integer, overflow traps	29	○					●	
Compile-time symbols; RAM-only patching	30, 31	○			●			
No class / perform: / thisContext	26			●	○			
Smaller syntax (no cascade, #( ), $a)	27							●

Row 25 (access modifiers parsed, not enforced) has no driver: it was never decided. Sources: each row's Why column; first-step design 2026-07-03 §1.

Figure 1. Each group of divergences from Newspeak against the reason its record gives. Numbers are rows of the dialect ledger [Ledger].

4.1 Kept

- **No global namespace.** Every name is reached through an object handed in: `platform List`, `platform space`. A class that was not given `platform` cannot reach the library at all (ledger rows 8, 9).
- **Modules are objects.** `class App packageUsing: manifest = (…) ( public main: platform args: args = (…) )`; the module is an instance parameterised by what it was given.
- **Nested classes and implicit-receiver sends**, with Newspeak's lookup order (lexical first), so an upgraded superclass cannot capture a name the enclosing module defines.
- **Slots with `::=` (mutable) and `=` (immutable), `::` assignment, keyword selectors, blocks, class-side bodies.**

4.2 Changed or removed

Area	Newspeak	nsl	Primary driver
Class references	virtual (late-bound slots)	static, resolved at compile time (row 1)	no JIT: production Newspeak makes virtual lookup free on Truffle/Graal
Composition	mixin application	single inheritance, one `delegate` (row 2)	as above; mixins are how virtual classes compose
Non-local return	`^` from any block unwinds to its home	compile error from an escaping closure (row 21)	simplicity: non-local return forced the reify/unwind/cross-fiber cascade
Concurrency	actors / processes	a run-to-completion reactor over a tuple space (row 22)	the same rewrite; determinism, predictable RAM
Exceptions	class hierarchy, resumable	`Record {#tag,#text}` values, predicate `on:do:`, `ensure:`, terminate-only (row 24)	errors cross a channel and the wire, so they must be values
Reflection	ambient	a `mirror` capability held only by the supervisor (rows 15, 16); no `class`, `perform:`, `thisContext` (row 26)	capability security; a closed-world image
Integers	unbounded	one type, ±(2^64−1), overflow traps (row 29)	one range for host, board and CBOR
Strings	collections of characters	immutable UTF-8 bytes, no `Character` (row 10)	one allocation, a GC leaf
Symbols	interned at run time	interned at compile time only (row 30)	the image is read-only
Methods	recompiled in place	read-only image; a debug build can override a method from RAM and restart its frame (row 31)	the image lives in flash
Syntax	cascades, literal arrays, `$a`, radix numbers	a 109-line grammar without them (row 27)	simplicity
Access modifiers	enforced	parsed, not enforced (row 25)	**undecided** — a finding of the 2026-10-06 audit

4.3 Why most of this is reversible

The structural half (no globals, modules as objects, confined privilege) was paid for first because it cannot be retrofitted, and most subtractions are additive to restore: static binding is the special case of late binding in which no class slot is overridden, so nothing written for nsl becomes wrong if virtual classes come back. The exceptions are rows 21–24: restoring non-local return, processes or resumption would bring back the machinery the rewrite deleted, and they are not on any plan.

5. Against Smalltalk-80 and Pharo

nsl's ancestry runs through Smalltalk — this system has been rewritten wholesale four times on the way here — and a Smalltalk programmer reads it easily. The differences are mostly about *where authority and state live*, not about expression.

	Smalltalk-80 / Pharo	nsl
Everything is an object; computation is message sends	yes	yes — including arithmetic, which falls back to a send off the integer fast path
Keyword selectors, blocks as closures	yes	yes
Control flow	`ifTrue:` etc. are messages, inlined by the compiler	the same, and the inlining is now the only implementation (row 19)
Namespace	one global `Smalltalk` dictionary; any method names any class	none; everything through `platform` (rows 8, 9)
The image	a live, mutable heap you develop in and ship	a frozen image built from source; the VM runs it from flash
`thisContext`, `become:`, `perform:`, `instVarAt:`	yes, ambient	no; frame access only through the confined mirror (rows 15, 26)
Processes and semaphores	yes	no; reactions that run to completion (row 22)
Non-local return	yes	only from inlined blocks (row 21)
Exceptions	class-based, resumable	value-based, predicate-filtered, terminate-only (row 24)
Class extension (adding methods to `Object`)	yes, from any package	no; the kernel is one module, and an app cannot reach its classes to extend them
Debugging	in-image, by anyone	over a line protocol to the confined debugger, from a host (§6)

What nsl takes from Newspeak rather than Smalltalk is the point of it: a module cannot reach anything it was not given, and that is what makes a capability *mean* something. What it gives up relative to Smalltalk is liveness: the system is developed on a host and deployed as an image, and hot patching is a debugging act, not the way programs are built.

6. Debug, patch, profile — on the device

A small device usually means printf and a reflash. nsl's answer is a set of tools that reach into the running VM over the device's one USB cable, through **custos**, the developer's peer on the host. Every tool is an HTTP call that answers in JSON first, with a web page and a terminal front over the same calls [Peer].

Tool	What it does today	Record
load	replaces the program without reflashing the firmware; the node reports the new program's hash (ESP32-C6)	[Image]
debug	breakpoints by source line; `step`, `next`, `finish`; the stack and named local variables at a stop	[Debugger]
patch	in a debug build, replaces one method of a program stopped at a breakpoint, restarts the frame, and continues into the new code — shown on an ESP32-C6 on 2026-10-06	[Patch]
profile	per-method call counts from the standard image on the ESP32-C6, so a field node is profiled without a reflash; it costs 1.3 % on the worst-case benchmark	[Profile]
log	the node's log and console, collected by custos	[Peer]

Every one of these names source, not addresses: the image map that ships beside each image gives every method's file and line, so a breakpoint, a profile sample or a crash lands on a line a person or an agent can open [Image].

Deep-sleep nodes — designed, not yet built. The same tools are designed to reach a node that is asleep most of the time, using the model §3 already gives: intent is a value the gateway holds. A breakpoint is written under the node's goals; the gateway holds it until the node next wakes and asks, whether that is seconds or two hours away. On a hit, the node reports the stop and stays awake while the debugger steps, and the liveness observer already knows not to reset a node held at a breakpoint [Peer]. A profile request works the same way: the next wake runs under the profiler and ships the counts [ProfileVerb]. Remote debugging of a sleepy node is the demonstration planned after version 0.1; until it runs, this paragraph is a design, not a result.

7. The case for an agent-first OO language

The Zephyr community is already using AI to write code, and does not trust it. Of 125 contributors surveyed, 54 % say it helps them write code faster — and 54 % say it gives unexpected or incorrect output they must fix frequently; 33 % spend more time validating its output than they would writing from scratch; 18 % say it helps them write code with fewer errors; and 78 % do not trust its suggestions without heavy review ([ZT10], Figure 35).

The problem those numbers describe is verification, and an agent verifies by running things and reading what comes back. So the platform is agent first: every custos verb has a machine-readable HTTP path before any human front [Peer], and the language should be chosen on the same terms. The argument, with what is measured and what is not:

1. **Keyword selectors name every argument at the call site.** `aSocket sendTo: host port: p data: bytes` cannot be called with the port and the host swapped without the code saying so; `sendto(fd, buf, len, 0, addr, alen)` can. The selector is the documentation, and it is the part of the code an agent pattern-matches on. *Not measured*: we have no count of argument-order errors by agents in nsl versus C or Python.

2. **The whole language fits in a context window.** Measured 2026-10-06 (bytes; tokens estimated at 3.5 bytes per token, not counted by a tokenizer):

Piece	Bytes	≈ tokens
Grammar (109 lines)	3 416	1 000
User's guide [Guide]	27 385	7 800
Dialect ledger [Ledger]	36 374	10 400
Kernel's public selectors (438 signatures)	6 397	1 800
Kernel source without comments	91 815	26 200
Kernel source as written	247 689	70 800

   Grammar, guide, ledger and the comment-free kernel come to about 160 KB, roughly 45 000 tokens: the complete language, standard library included, fits a 200 k window with room for the program, and the grammar plus guide plus public selectors fit easily in the 64 k window of a local model. A Python or C agent works from memory of a library it cannot see; an nsl agent can be given all of it.

3. **Authority is visible in signatures.** A module that can use the network says so in its header (`usingPlatform: platform net: net`); a channel capability can only narrow. An agent — or a reviewer of an agent's work — can read what a piece of code is able to touch without reading its body. (Weakened today: `platform` hands out every device; see §8.)

4. **A small, closed primitive set.** 160 primitives, all other code is nsl; there is no FFI to wander into (§2).

5. **Errors written for a reader that is an agent.** A message not understood names the receiver's class and the nearest selector it does understand; about 1 KB of flash is spent on that on purpose, because "an operator that is an agent is exactly as good as the error line it is fed".
6. **Every artefact is addressable**, and every tool answers in data (§6).
7. **The loop is short.** A program runs first on the VM on the host, then is loaded onto the device without a reflash; a stopped method can be patched and resumed. A short loop makes each check cheap, and cheap checks are what the 33 % above are missing. *Not measured*: we have not timed the loop end to end.

Against it, honestly. Dynamic typing gives an agent nothing at compile time: a misspelt selector is found when it runs, which is why item 5 exists. nsl is not in any model's training data, so an agent writes it from the context it is given, not from habit — which is the bet item 2 makes and has not yet tested. What would test it: the same small board tasks given to an agent in nsl (with the language in context) and in Zephyr C, scored on first-run correctness. After version 0.1 that comparison should be shown, not told.

8. Costs, and what would lift each

Cost	Felt as	Would lift it	Ledger row
No virtual classes or mixins	a library cannot be specialised per target except by choosing a module at boot	restore late-bound class slots on the same vtable substrate	1, 2, 6
No non-local return	loops that would `^` out of a `do:` are rewritten as `detect:ifNone:` or a flag	not planned	21
No processes	a long computation starves timers	split it across reactions; the liveness observer shows the stall	22
No resumable exceptions	a handler cannot supply a value and continue at the raise	`on:do:` answering the default	24
Per-byte work is interpreted	the CBOR codec is nsl, and every byte it reads costs a method activation	move the hot methods to C by native substitution (§2)	—
`platform` is all-powerful	any app holds every device	narrowed platforms, or `serial:`/`net:`-style header grants for devices	32
Access modifiers unenforced	`private` documents intent and nothing more	emit them and check, or delete the check	25
Hot patch limited to known constants	a fix that names a new selector or string cannot be patched	intern in RAM; carry offsets in the image map	30, 31
No `Character`, byte strings	text is bytes	a rune-aware protocol	10, 12

9. Open questions

- Row 25: enforce access modifiers or delete the mechanism?
- Row 32: is `platform` meant to hand out every device, or should devices become header grants as `serial:` and `net:` are?

- §7: is the agent-in-nsl versus agent-in-C comparison worth running before 0.1 is public, so that the claim is measured rather than argued?

Appendix A. A Thread node in 36 lines

A node that joins the gateway over Thread, checks in every two seconds, publishes a log line each time, and answers a name the gateway gives it. Thread itself is attached by the firmware before the VM starts; the program sees an IPv6 UDP socket. It compiles for the nRF52840; it is cut down from the bench's porta node, which runs, and has not itself been run [Example].

```
class ThreadSensorNode usingPlatform: platform net: net = (
  | space bridge timers ticks ::= 0. |
) (
  gwAddr  = ( ^ 'fdc5:9109:9afe:97ea:9541:d963:f8d9:8567' )   (* the gateway, mesh-local *)
  gwPort  = ( ^ 6969 )
  cadence = ( ^ 2000 )                                         (* ms between check-ins *)

  public main: platform2 args: args = (
    | sock tickChan |
    space:: platform space.
    sock:: net udpBindPort: 16970.
    sock == nil ifTrue: [ 'bind failed' println. ^ 1 ].
    bridge:: platform PortaBridge
      socket: sock space: space id: platform nodeId peer: self gwAddr port: self gwPort
      spans: platform Spans new.
    bridge subscribeStream: 'tel/log'.
    bridge liveness: self cadence misses: 16.
    bridge publishBuild: #nrf52840.
    (space channel: 'goal/name') whenever: '' do: [ :v | space put: 'obs/name' value: v ].
    timers:: platform timers.
    space whenever: 'sys/timer' do: [ :e | timers dispatch: e ].
    tickChan:: space channel: 'sys/tick'.
    tickChan whenever: '' do: [ :m | self tick ].
    bridge startTicking: timers on: tickChan.
    [ true ] whileTrue: [ platform pumpEvents. space drain ].
    ^ 0
  )

  tick = (
    ticks:: ticks + 1.
    bridge checkIn.
    space out: 'tel/log' value: (platform Record tag: #Log
      with: #lvl -> #info with: #txt -> ('tick ' , ticks printString)).
    ^ self
  )
) : (
  public packageUsing: manifest = ( ^ self new )
)
```

The header says what the program may touch: the platform and the network, nothing else. The gateway sees the node's log, its build and its health; writing `goal/name` at the gateway reaches the node at its next check-in, and the answer comes back at `obs/name`.

References

- **[ZT10]** B. Trinkenreich, A. Lawson and H. Carter, *Zephyr Turns 10: A Decade of Adoption, Maturity, and Ecosystem Evolution*, The Linux Foundation, March 2026, CC BY-ND 4.0 — copy in this repository. 900+ boards;

the learning curve; Figure 35 (p. 30), GenAI.

- **[OEOM]** I. Piumarta and A. Warth, *Open, Extensible Object Models* (2007).
- G. Bracha et al., *Modules as Objects in Newspeak* (ECOOP 2010); G. Bracha and D. Ungar, *Mirrors: Design Principles for Meta-level Facilities of Object-Oriented Programming Languages* (OOPSLA 2004); A. Goldberg and D. Robson, *Smalltalk-80: The Language and its Implementation* (1983).
- **[Ledger]** The nsl dialect ledger: every divergence from Newspeak, its reason and what would revisit it (rows 1–32, audited 2026-10-06).
- **[Guide]** Programming with nsl, the user's guide.
- **[VM]** oevm — what the VM is.
- **[Pivot]** The July 2026 rewrite's design: no non-local return, no fibers, the reactor; the 256 KB target.
- **[MA]**, **[MD]** Milestone A (what the rewrite deleted, measured) and Milestone D (the supervisor).
- **[Boards]** The board profiles in `run.sh` and the board ports; the nRF52840 RAM split from that build's linker symbols.
- **[RP2350]** First numbers on the RP2350.
- **[Image]** The image and its map.
- **[Porta]** porta's architecture, §5: power modes and app lifecycle.
- **[Serial]** Adding the serial device: a device target is purely additive.
- **[Hanoi]** Native substitution, the Hanoi spike.
- **[Liveness]** The liveness observer.
- **[Peer]** custos, the developer's peer: agent first (D11–D13), remote debugging of a sleepy node (D9).
- **[Debugger]** Tutorial 4, the debugger.
- **[Patch]** Stop, patch, restart, resume.
- **[Profile]** Profiling over USB.
- **[ProfileVerb]** The profile verb: profiling a deep-sleep node on its next wake.
- **[Example]** `ThreadSensorNode.nsl`, cut down from `PortaBridgeNodeNrf.nsl`.